BUBT Class Routine & Scanner Engine

Official Technical Guide & System Specification

This document provides a comprehensive guide to the **Class Routine System and 2D Spatial OCR Scanner Engine** implemented in the **CRConnect** application for Bangladesh University of Business and Technology (BUBT).

System Overview

The CRConnect Routine Engine digitizes paper routines, PDF screenshots, and annex web portal schedules into interactive, real-time academic schedules for BUBT students and teachers.

Key Capabilities

- ■ **Camera & Gallery OCR Scanner**: Point the camera or upload a screenshot to instantly parse full weekly routines.
- ■ **2D Cell-Bucket Matrix Engine**: High-precision spatial mapping ensuring zero cross-column or cross-row text bleeding.
- ■ **Dual Synchronization**: Local SharedPreferences persistence combined with real-time Firebase Cloud Firestore cloud sync (`routines/{dept}/{intake}/{section}/schedule`).
- ■■ **Routine Management**: Includes full routine deletion, manual class addition, and daily schedule filtering.
- ■ **Up-Next Countdown & Attendance**: Real-time status banners, live countdowns, and attendance markers (`Attended` / `Missed`).

BUBT Routine Grid Structure

BUBT routines follow a standardized weekly grid layout:

1. Days of the Week (7 Rows)

Short Code	Full Day Name
`SAT`	Saturday
`SUN`	Sunday
`MON`	Monday

`TUE`	Tuesday
`WED`	Wednesday
`THR` / `THU`	Thursday
`FRI`	Friday

2. Time Slots (6 Columns)

Slot Index	Time Interval
Slot 0	`08:15 AM - 09:45 AM`
Slot 1	`09:45 AM - 11:15 AM`
Slot 2	`11:15 AM - 12:45 PM`
Slot 3	`01:15 PM - 02:45 PM`
Slot 4	`02:45 PM - 04:15 PM`
Slot 5	`04:15 PM - 05:45 PM`

3. Cell Data Elements

Inside each grid cell, the following four items are parsed:

- **Course Code**: e.g., `CSE 210`, `CSE 207`, `CSE 341`, `CSE 208`, `CSE 342`, `CSE 209`
- **Faculty Initial (`FC`)**: e.g., `FC: MDSS`, `FC: AAG`, `FC: MJ`
- **Room Number (`R`)**: e.g., `R: 2517`, `R: 2810`, `R: 2808`, `R: 2419`, `R: 2318`
- **Intake & Section**: e.g., `Intake: 55 - 6`

Scanner Engine Architecture

The OCR parser is built on top of **Google ML Kit Text Recognition** and utilizes a multi-stage spatial algorithm for 100% recognition accuracy:

1. Atomic Element Decomposition

ML Kit sometimes merges text from adjacent table cells into a single horizontal line (e.g., `"CSE 210 CSE 210"` across Column 1 and Column 2).

- The parser inspects individual `Text.Element` word groups.
- Multiple course matches in a line are split into atomic word units with their own unmerged bounding boxes and exact X-coordinates.

2. Header Anchor Extraction & Deduplication

- **Day Anchors**: Identified from day names (`SAT`–`FRI`).
- **Time Slot Anchors**: Deduped so each of the 6 time slots (`08:15`, `09:45`, `11:15`, `01:15`, `02:45`, `04:15`) has **exactly one** unique anchor X-coordinate corresponding to the start of its header cell.

3. Grid Midpoint Binning Algorithm

Rather than using arbitrary pixel radiuses, column boundaries are calculated using exact midpoints between adjacent column centers:

$$\text{Midpoint}_i = \frac{\text{Anchor}_i + \text{Anchor}_{i+1}}{2}$$

Any text item with center X-coordinate falling between Midpoint_{i-1} and Midpoint_i is guaranteed to land in **Time Slot i**. Row boundaries are computed similarly along the Y-axis for days of the week.

4. Cell-Bucket Text Grouping

All text lines within a specific `(DayRow, TimeColumn)` grid box are grouped into a dedicated **Cell Bucket**.

- Faculty initials (`AAG`, `MJ`, `MDSS`) and Room numbers (`2517`, `2810`, `2808`, `2419`, `2318`) are extracted **exclusively** from items inside that exact cell bucket.
- Cross-column and cross-row line contamination is mathematically impossible (0% bleed).

Regex & Pattern Rules

Feature	Regex Pattern	Example Match	Extracted Value													
Course Code	`(?i)((?:CSE\	MAT\	ENG\	EEE\	PHY\	CHEM\	BUS\	STA\	ACT\	FIN\	LAW\	GED\	TEX\	CIV\	[A-Z]{2,4})[-_]*[0-9]{3,4}[A-Z]?)`	`CSE210`	`CSE210`
Faculty Initial	`(?i)(?:FC\	Faculty\	Instructor\	Teacher)[:\s.-]*([A-Za-z]{2,5})\b`	`FC: MDSS`	`MDSS`											
Room Number	`(?i)(?:R\	Room\	Rm)[:\s.-]*([0-9]{3,4}\	Lab[^0-9]*[0-9]+)\b`	`R: 2517`	`2517`											
Intake / Sec	`(?i)(Intake\	Int\	Sec\	Section)[:]*([0-9]{1,3})[_/,:-]*([0-9]{1,3})`	`Intake: 55-6`	Intake `55`, Sec `6`											

Candidate Blacklist Protection

To prevent labels from being misidentified as faculty initials, the following words are strictly blacklisted:

`FC`, `FACULTY`, `TEACHER`, `INSTRUCTOR`, `R`, `RM`, `ROOM`, `LAB`, `SAT–FRI`, `INTAKE`, `SEC`, `SECTION`, `CLASS`, `DAY`, `TIME`, `BUILDING`, `BLDG`, and department codes (`CSE`, `MAT`, `EEE`, etc.).

Dual Storage & Real-Time Sync

1. **Local Disk Storage**:
   • `RoutineRepository` (`crconnect_routine_preferences` SharedPreferences) for Jetpack Compose `RoutineTab`.
   • `RoutineLocalCache` (`bubt_routine_local_cache` SharedPreferences) for Java `RoutineActivity`.
2. **Cloud Firestore Sync**:
   • Path: `routines/{deptKey}/{intakeKey}/{secKey}/schedule`
   • When a routine is scanned, old documents at this path are cleared and new scanned class entries are written via an atomic `WriteBatch`.
   • When offline, the app seamlessly falls back to the locally saved routine.

Key Files & Implementation Locations

File Path	Description
[student/RoutineOcrParser.kt](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/student/RoutineOcrParser.kt)	Core Kotlin 2D Spatial OCR Parser engine for Compose UI
[RoutineActivity.java](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/RoutineActivity.java)	Native Java Routine Activity & OCR parser
[student/RoutineRepository.kt](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/student/RoutineRepository.kt)	SharedPreferences local persistence manager
[RoutineLocalCache.java](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/RoutineLocalCache.java)	Java local caching utility
[student/StudentPortalViewModel.kt](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/student/StudentPortalViewModel.kt)	ViewModel managing state, local storage, and Firestore real-time sync

[student/RoutineTab.kt](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/student/RoutineTab.kt)	Jetpack Compose UI for Routine Schedule view and Bottom Sheet options
[RoutineCameraActivity.kt](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/java/com/example/crconnect/RoutineCameraActivity.kt)	CameraX capture activity with flash and flip controls
[dialog_scan_routine_options.xml](file:///C:/Users/shadd/StudioProjects/CRConnect-App/app/src/main/res/layout/dialog_scan_routine_options.xml)	Bottom Sheet layout for camera scan, gallery pick, and routine deletion

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Best Practices for Scanning Routines

- **Lighting**: Ensure adequate lighting without heavy screen glare or flash reflection.
- **Framing**: Capture the full routine table including the top header row (`08:15 AM to ...`) and left day column (SAT–FRI).
- **Format**: Screenshots directly from PDFs or BUBT Annex portal (`annex.bubt.edu.bd`) produce 100% optimal results.